

Customer Shopping Behavior Analytics Report

1. Project Overview

This project analyzes retail customer shopping behavior using Python, SQL, and Power BI. The objective is to identify purchasing trends, customer segments, and business opportunities that can help organizations improve revenue and customer engagement.

2. Business Problem

Retail companies collect large amounts of customer transaction data but often struggle to transform it into actionable insights.

The goal of this project is to:

- Understand customer purchasing patterns
- Analyze spending behavior
- Identify high-performing products and categories
- Evaluate subscription and loyalty trends
- Support business decision-making through data visualization

3. Tools & Technologies

- Python (Pandas, NumPy)
- SQL
- Power BI
- Google Colab

4. Data Preparation

The dataset was cleaned and transformed using Python.

Activities performed:

- Missing value handling
- Data type conversion
- Feature engineering
- Customer segmentation preparation
- Data validation and quality checks

5. SQL Business Analysis

The following business questions were analyzed:

- Revenue by gender
- Average purchase amount by gender
- Top-rated products
- Shipping type analysis
- Subscription impact on revenue
- Discount usage analysis
- Customer segmentation
- Repeat buyer behavior
- Revenue contribution by age group

6. Dashboard Insights

Key findings:

- Clothing category generated the highest revenue.
- Young Adult customers contributed the largest share of sales.
- Subscribers showed higher overall engagement.
- Repeat buyers demonstrated stronger subscription adoption.
- Certain shipping methods were associated with higher purchase values.

7. Business Recommendations

1. Increase marketing efforts toward high-value customer segments.
2. Promote subscription programs to improve customer retention.
3. Focus inventory planning on top-performing categories.
4. Introduce targeted offers for repeat customers.
5. Use dashboard monitoring for continuous business tracking.

8. Conclusion

This project demonstrates an end-to-end analytics workflow including data preparation, SQL-based business analysis, dashboard creation, and business recommendations. The insights generated can help organizations make data-driven decisions and improve customer experience.